

MAGNET RUNDOWN UNIT

Primary Magnet Run Down Unit

GE Part number **46-294231G1**

Operating and Service Manual

(46-318393 Revision 11)

17 June 2015

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46-318393

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Revision 11

Revision History

REV	Affected Page(s)		Description of Content Changed	Reason for Change or Change Control Number	Author	Effective date of Document	Periodic review date
11	ALL		Revised per HII review	ECO	Don Gorg	My Workshop Release	Does not Apply to Service Manuals
Yes	No	NA		Enter Validation Information or Rationale if "NO" or "NA" for Validation or Training			
☐	☒		Typo or Format Change				
☐	☒	☐	Regulatory Filing				
☒	☐	☐	Validation	As noted within Summary Table			
☐	☒	☐	Training	No change to existing techniques.			
☒	☐	☐	Communication	Upon release.			
☒	☐	☐	Labeling, (list)	Document is a Service Note (classified as Labeling)			
☐	☒	☐	Other Documents, (list)				
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Change Summary Table

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1.0 PREINSTALLATION

1.01 Each MRU is shipped with an operating manual and service manual. The batteries are installed and were fully charged before leaving the factory. The charge on the batteries should be verified at 20 volts (minimum) before any adjustments are made.

1.02 Verify that AC power of 115V/230V (nominal), 50-60Hz, 35 watts (maximum) is available within 6 ft. of the mounting location. MRU's are shipped strapped for 115V.

1.03 Inspect mounting surface and obtain the type of mounting fasteners required for the surface involved. Use fasteners capable of supporting a 32 lb. (14.5kg) load. The case is drilled for 1/4" or 6 mm bolts arranged in a rectangular pattern. Vertical spacing is 5.50 inches and horizontal spacing is 10.00 inches. Use flat washers under the fasteners.

1.04 ALL 200-240 volt applications MUST be PERMANENTLY INSTALLED. 100-120 volt applications may be PERMANENTLY INSTALLED if desired. For any PERMANENTLY INSTALLED application, refer to the SERVICE MANUAL schematic for connections to TB1. All PERMANENT installations shall be connected to a dedicated Mains circuit, with a disconnect device that is in compliance with local regulatory standards. If conduit is to be used, contact building maintenance or an outside contractor as required to have conduit/wiremold installed.

1.05 Verify that magnetic field intensity is less than 200 Gauss at the mounting location.

NOTE: Confirm fuse installed in location F1 is correct rating for selecting AC incoming voltage, see schematic section for Jumper locations and Fuse values.

1.06 A No. 2 Phillips screwdriver is required to open the front cover.

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2.0 INSTALLATION AND ADJUSTMENT

WARNING: PREVENT INADVERTENT MAGNET QUENCH BY ENSURING CABLE P2 TO MAGNET HEATERS ARE NOT CONNECTED TO MRU J2.

2.01 Mount the MRU to the wall surface. Connect AC power.

NOTE: Confirm fuse installed in location F1 is correct rating for selecting AC incoming voltage, see schematic section for Jumper locations and Fuse values.

2.02 Connect a DVM, positive to the test point labeled +20.7V and the negative to the test point marked ground. The multimeter should read 20.7 VDC. Adjust R3 if necessary to obtain 20.7 V. Verify CHARGER POWER LED on the front panel is lit.

2.03 Remove the 20 ohm 5% ceramic resistor from the plastic bag and connect it between lugs 1 and 2 of TB2.

WARNING: THE 20 OHM TEST RESISTOR WILL BECOME HOT DURING THIS TEST.

2.04 Install fuse F2 (2.5A T 5x20mm fuse). Measure the voltage at test point +20.7V with respect to ground; it should be between 20.1 and 20.7 volts. If the MRU batteries have been stored for more than one year without charging, it may be necessary to wait for up to 8 hours for the voltage to reach 20.7 volts.

2.05 Depress and release the RUN DOWN button. Measure the elapsed time until the RUN DOWN push-button pops back out. The time must be a minimum of 30 seconds. The red HEATER ACTUATED LED will light and remain lit until reset with switch S6. Press push-button switch S6 on the printed circuit board to reset the HEATER ACTUATED LED.

2.06 Remove the test resistor from TB2. Disconnect the DVM and close the front cover.

2.07 Verify the green CHARGER POWER LED is lit.

2.08 Depress and hold the TEST BATTERY switch for 15 seconds and the green BATTERY TEST LED should light. The set point is 16.5 volts.

WARNING: BEFORE CONNECTING P2 TO J2, ASSURE THAT AC POWER HAS BEEN APPLIED TO THE MRU FOR AT LEAST 3 MINUTES AND THAT THE RUNDOWN BUTTON HAS NOT BEEN ACTUATED FOR AT LEAST 3 MINUTES AND IS NOT DEPRESSED. FAILURE TO HEED THIS WARNING MAY CAUSE A MAGNET QUENCH.

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2.09 Open the cover and connect the main coil heater cable plug P2 to matching connector J2. Close and fasten the cover. Depress the TEST HEATER switch. The green HEATER TEST LED should light.

2.10 Complete the maintenance log with date and signature.

3.0 WEEKLY TESTING

WARNING: POTENTIAL MAGNET QUENCH

Do not press the Rundown/Quench button throughout this process

Do not reconnect MRU cable if it is found disconnected. GEHC strongly recommends that you stop using the system, and immediately call your qualified service representative

Confirm that the MRU is connected to the magnet and operating properly by performing the following test on the MRU.

3.01 Verify the green CHARGER POWER LED is lit.

3.02 Depress and hold the TEST BATTERY switch for 15 seconds. The green BATTERY TEST LED should light and remain lit while the TEST BATTERY switch is depressed.

3.03 Place the TEST HEATER toggle switch in the A position. The green HEATER TEST LED should light and remain lit until toggle released.

Note: Wait at least 5 seconds before switching between position A and B

Not doing so may cause the red Heater Actuate LED to light

3.04 Place the TEST HEATER toggle switch in the B position. Green HEATER TEST LED should light and remain lit until toggle released.

If the MRU test does not perform as described in each of the 4 steps above, with the specified LED lighting in each step, GEHC strongly recommends that you stop using the system, and immediately call your qualified service representative.

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4.0 THEORY OF OPERATION

4.01 General System Description

The Model 46-294231G1 MRU is designed to provide a timed (30seconds minimum, 50 seconds nominal) 800 mA DC current source to both of the heaters in a 1.5T magnet system. In addition, test circuits are provided to verify battery charger operation, battery charge condition and heater resistance limits.

4.02 AC power supply and regulator

115 VAC is permanently connected through fuse F1 to T1. A full wave rectifier CR4 and C1 provide filtered DC to U1, an LM350T IC regulator. U1 has internal short circuit and over-temperature limiting. Diodes CR2 and CR1 provide transient and reverse polarity protection. R3 is used to set the float charge on the battery assembly to 20.7 volts. DS1 provides a continuous indication of regulator output with isolation provided by diode CR3 to prevent lamp indication with AC input or regulator failure. During the BATTERY TEST function, a 15 ohm test load is placed across the battery and the regulator is disabled by the second half of S1 to ensure that the battery is the only source of power.

4.03 Test Circuits

During the BATTERY TEST mode S1 also applies an input via R27 to a comparator circuit consisting of 1/4 of U4 and associated components. U5 provides a constant current source to a voltage divider to establish a 16.5 volt trip point reference. When R26 is properly set, LED DS2 (BATTERY TEST) will light at 16.5 volts and above and not light below 16.5 volts. Resistors R29, R30 and capacitor C7 provide a low pass filter and transient protection to the comparator input.

Two other sections of U4 are used to sense upper and lower limit heater resistance during the HEATER TEST mode. Pressing S2 to either the A or B position allows approximately 5mA to flow from the battery through R5 and the heater element under test.

The resulting voltage drop across R5 is sensed via R16 at two comparators. R8 sets a 12 ohm lower limit and R20 a 32 ohm upper limit for heater resistance. If the heater resistance is between 12 ohms and 32 ohms, the HEATER TEST LED will light. Since this is a critical test, a switch is provided to TEST HEATER LED, S5. U2 and U3 provide constant current sources for stable reference voltage dividers. R12 through R15, C4 and C5 provide low pass filtering and transient protection.

4.04 Time Delay and Output Circuits

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Actuation of the covered push-button RUN DOWN switch S4, energizes both heaters by resetting a thermally timed circuit breaker S4. The 100mA rated sensing element passes approximately 170 mA for a nominal 50 seconds at which time the circuit breaker opens, interrupting current flow through the heaters. The power supplied to the circuit breaker heating element is limited by resistor R34. The red LED marked HEATER ACTUATED will light and stay lit whenever the RUN DOWN switch is activated, or if the HEATER TEST switch is activated and the heater circuits are open. The latching circuit should be reset only by General Electric Field Service personnel. S6, a momentary push-button switch is located on the main PC board and resets the SCR latching circuit.

ACCIDENTAL HEATER ACTIVATION WILL RESULT IN LOSS OF EXPENSIVE COOLING FLUIDS AND MAY CAUSE PERMANENT DAMAGE TO THE 1.5TMAGNET.

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5.0 SCHEMATICS, FRAME ASSEMBLY, WIRING HARNESS

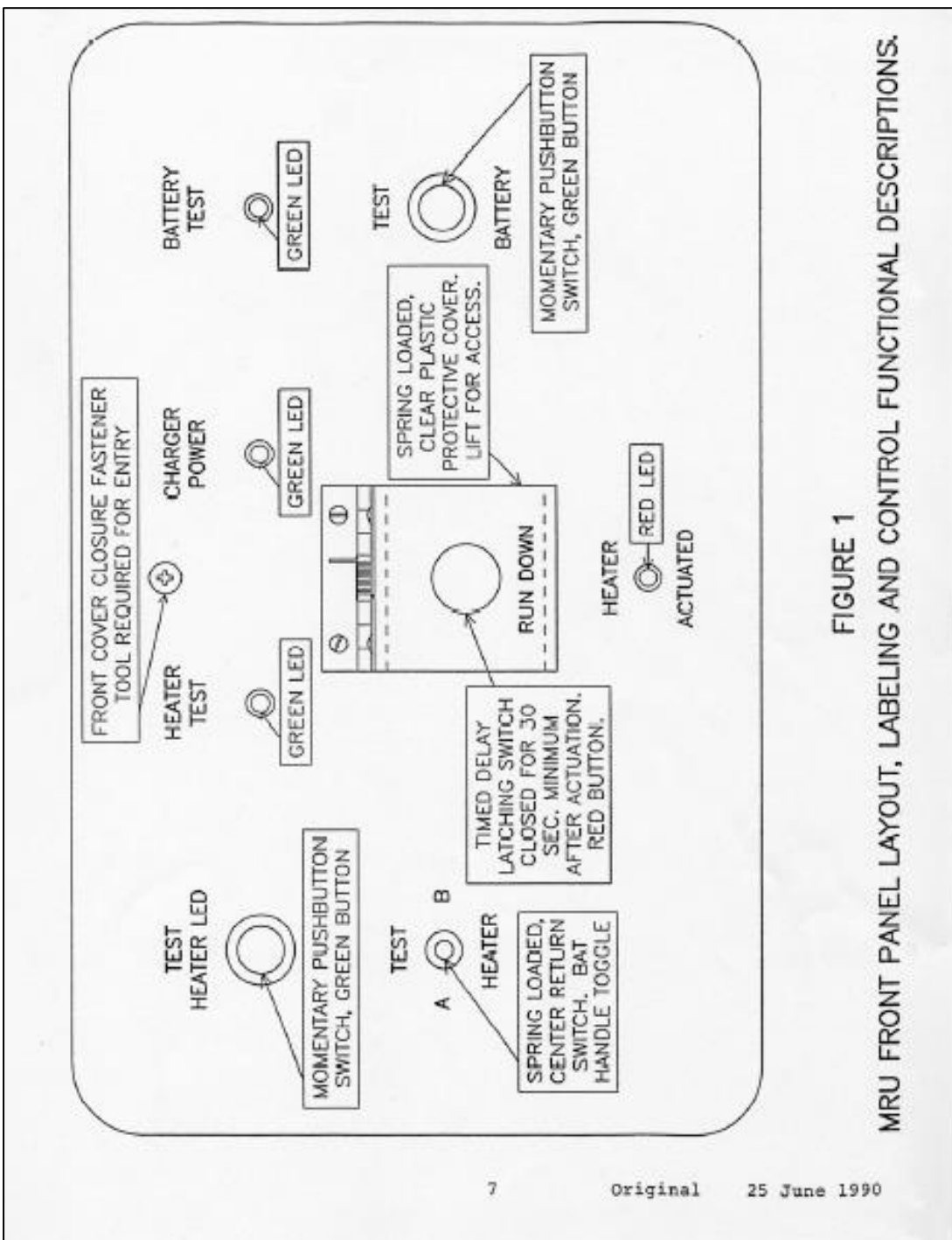


FIGURE 1
MRU FRONT PANEL LAYOUT, LABELING AND CONTROL FUNCTIONAL DESCRIPTIONS.

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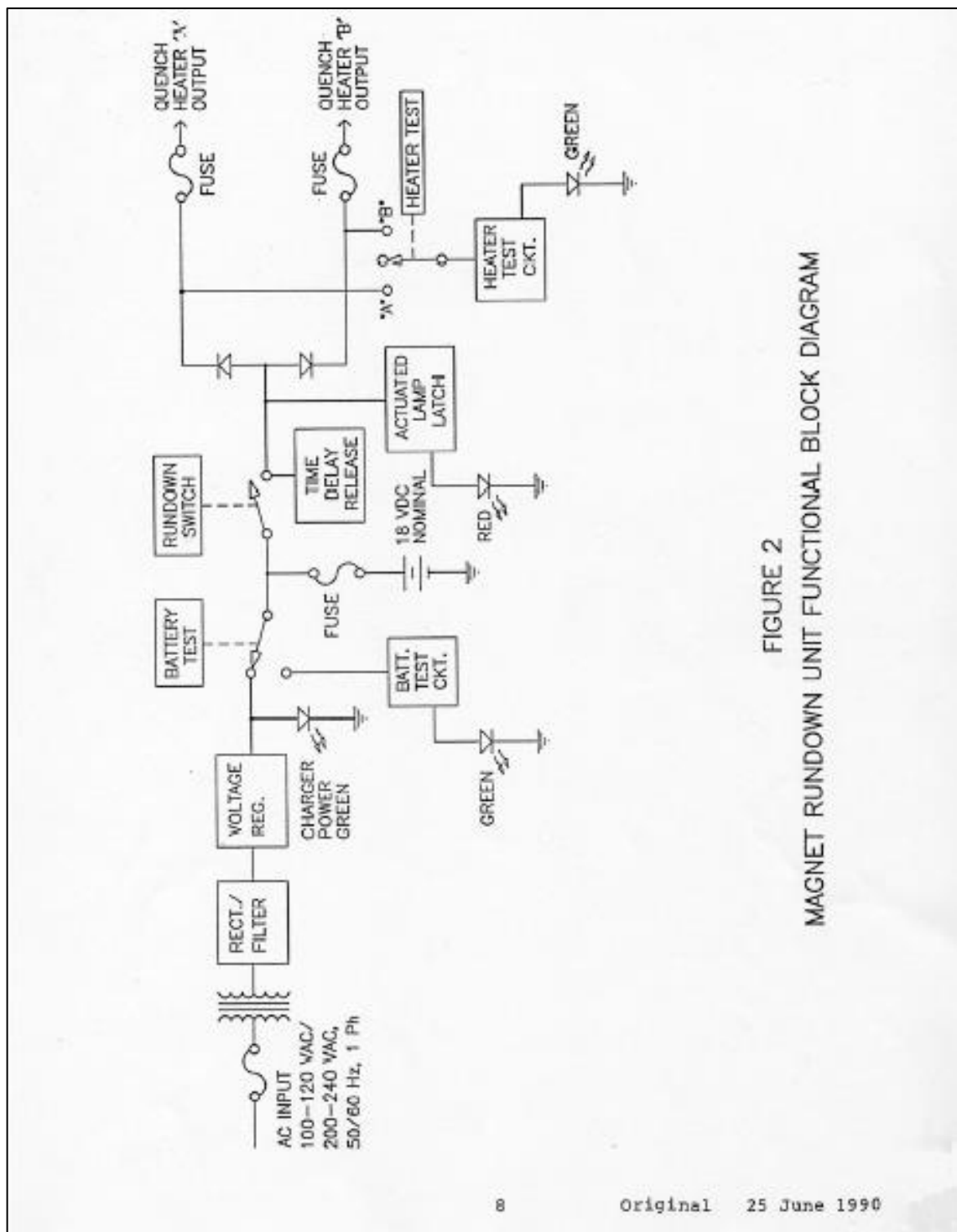


FIGURE 2
MAGNET RUNDOWN UNIT FUNCTIONAL BLOCK DIAGRAM

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Original 25 June 1990

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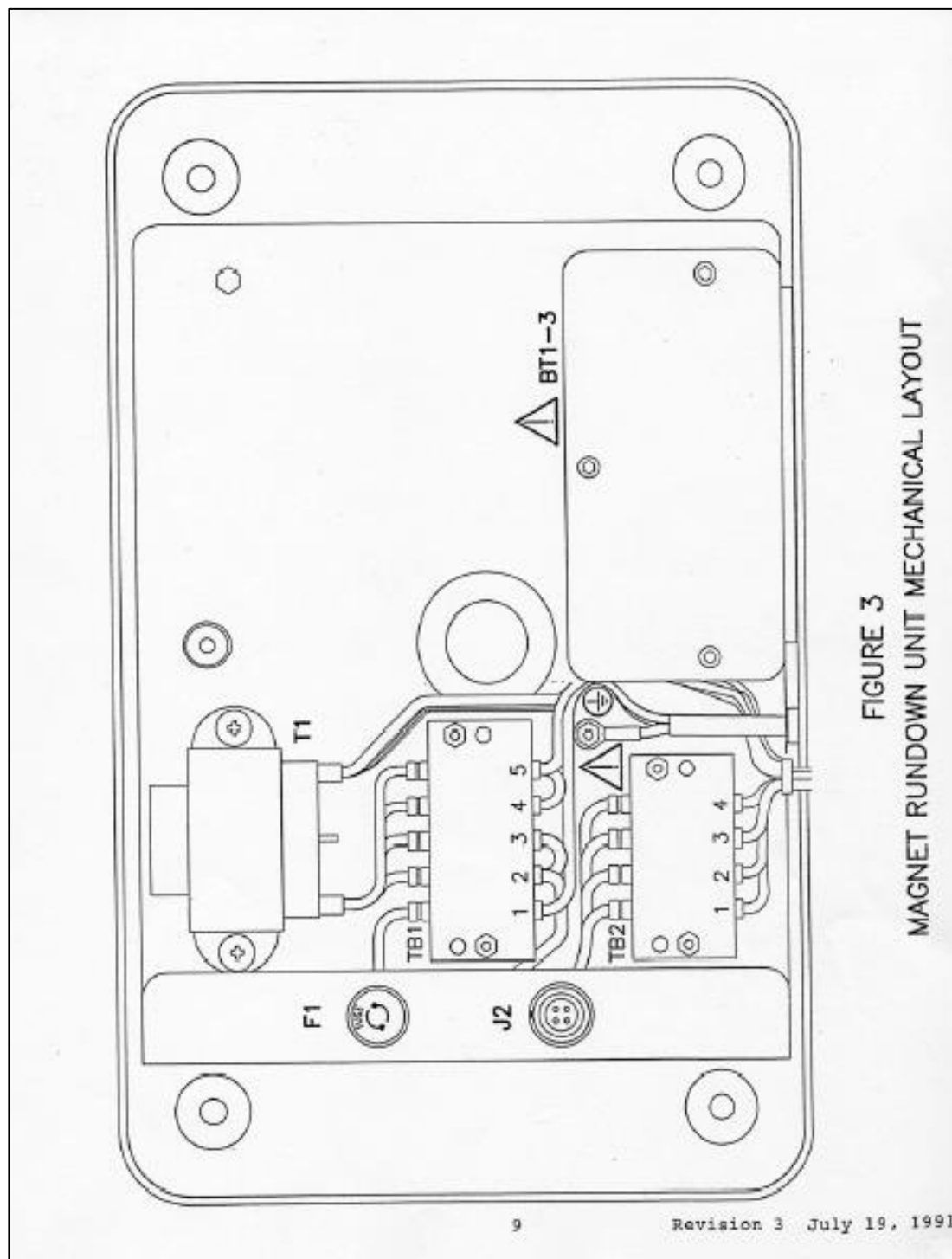


FIGURE 3
MAGNET RUN DOWN UNIT MECHANICAL LAYOUT

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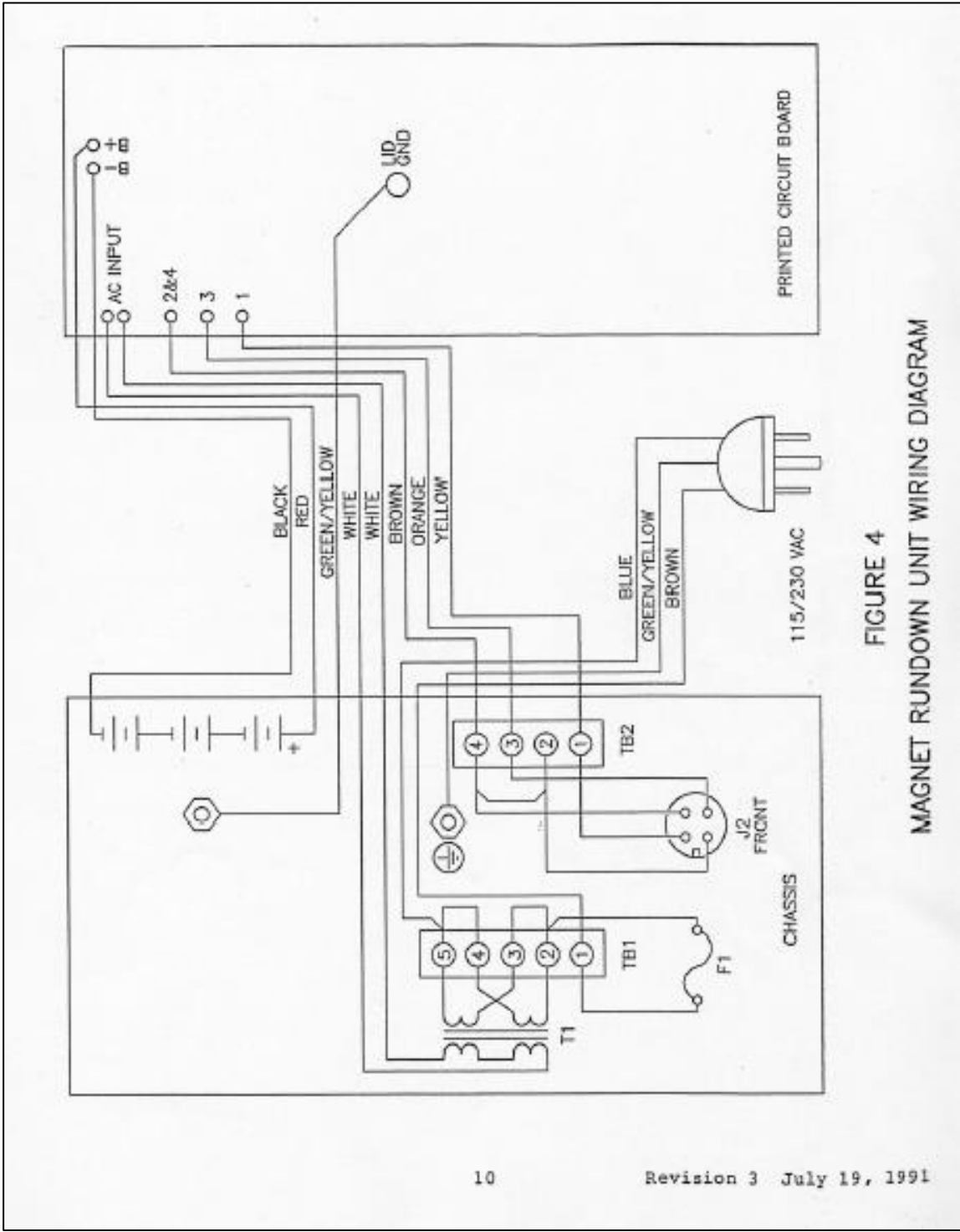
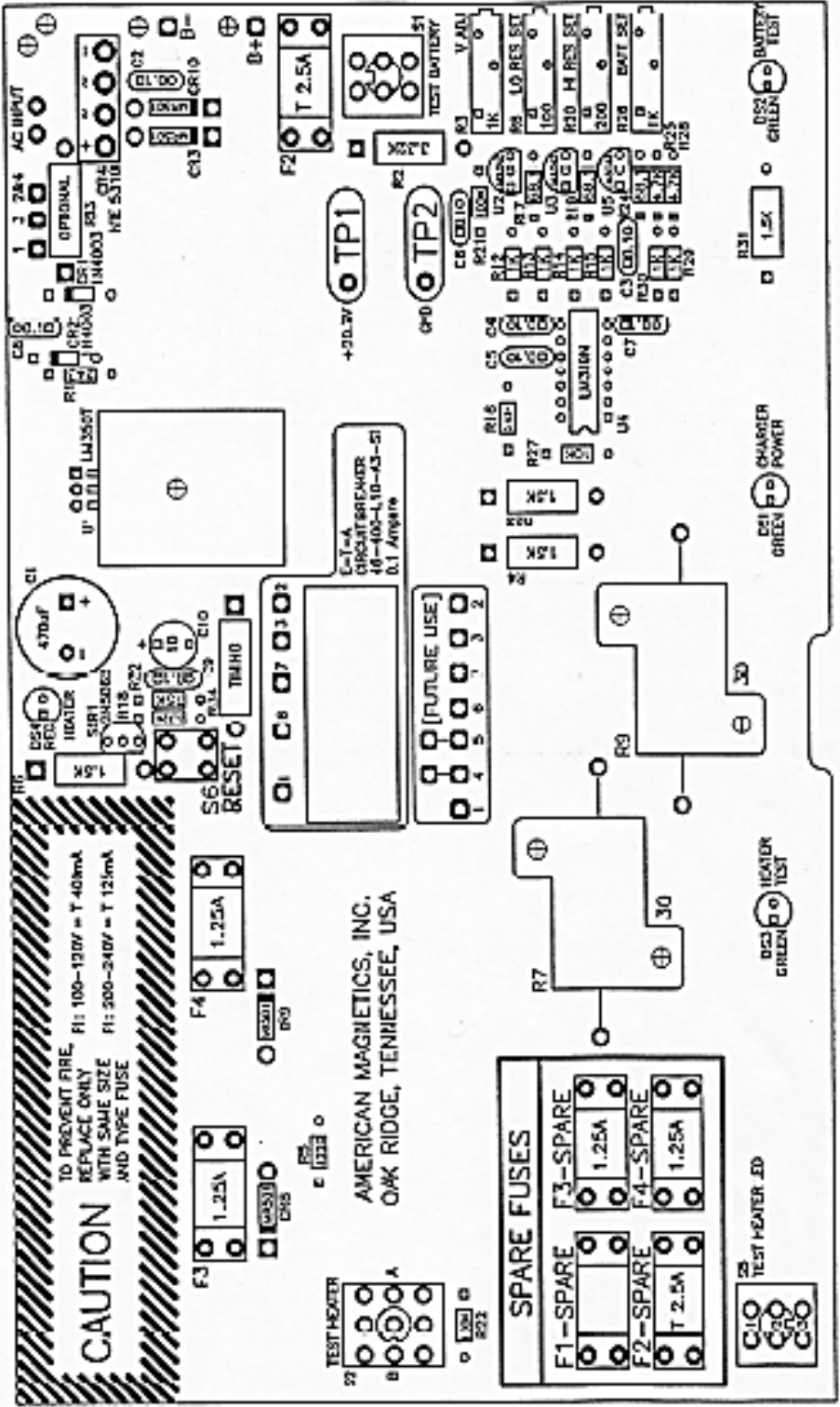


FIGURE 4
MAGNET RUNDOWN UNIT WIRING DIAGRAM

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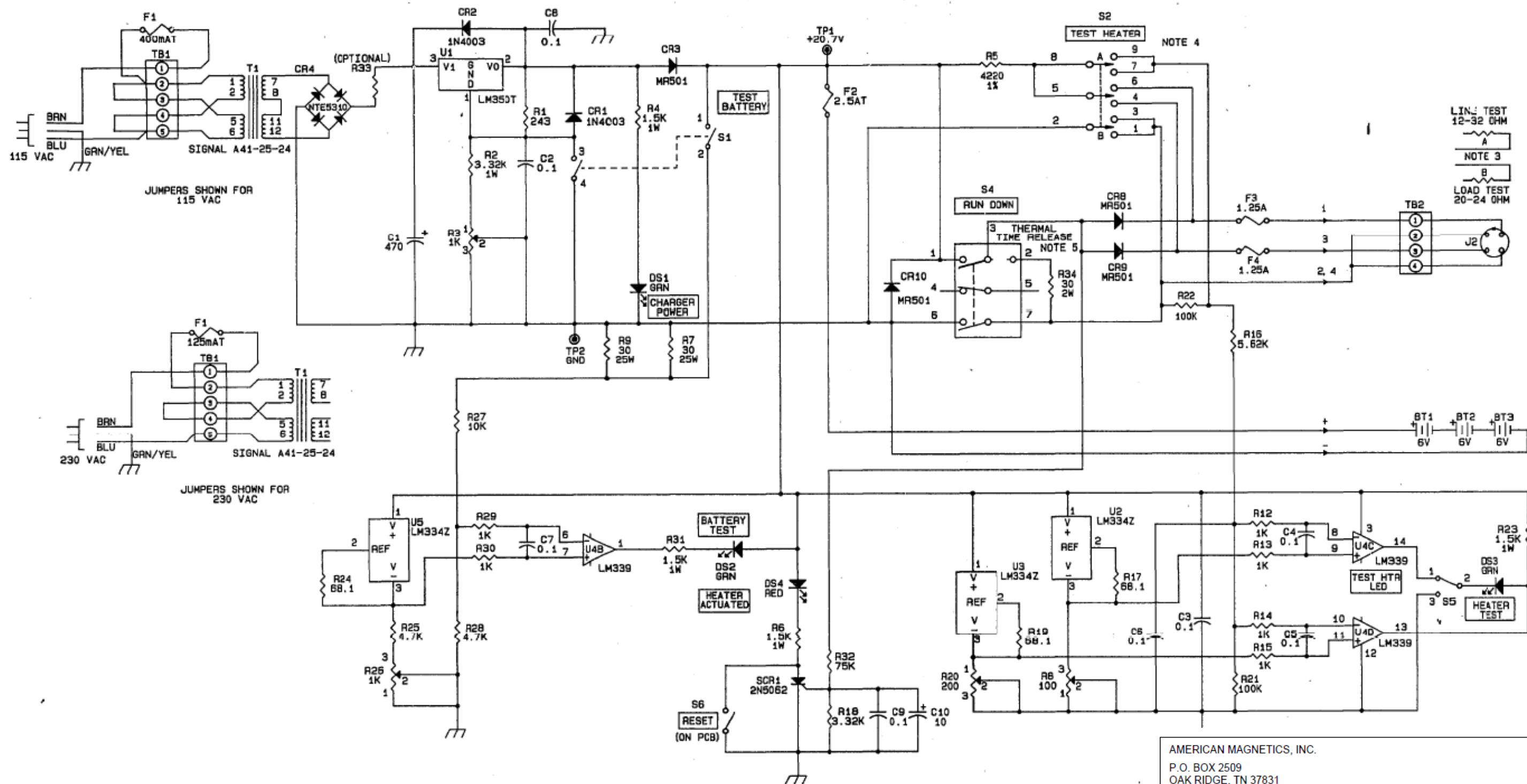
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AMERICAN MAGNETICS, INC. P.O. BOX 2509 OAK RIDGE, TN 37831 (423) 482-1056		
Title GE MAGNET RUNDOWN UNIT SCHEMATIC		
Size C	Document Number 90-2567-1494	Rev 4
Date: Monday, October 23, 2000	Sheet 1	of 1

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6.0 REPLACEMENT PARTS LIST

GE part number	Description	Qty
46-294231P5	Battery, Gel Cel 6V 1.2AH Yuasa NP1.2-6	3
5363631	Kit,Spare Fuse for MRU	1

NOTE: The if the original (46-294231P5) Battery, Gel Cel 6V 1.2AH Yuasa NP1.2-6 is obsolete, then use replacement (5199490) Battery Pack Asm, MRU2006, per ECR2128752.

7. PERIODIC MAINTENANCE

WARNING: HAZARDOUS VOLTAGES ARE ACCESSIBLE AT COMPONENTS F1, TB1, S3, RV1, AND RV2 AND AT THE LOWER TERMINALS OF T1. TO AVOID ELECTRIC SHOCK, USE CAUTION WHEN THE MRU COVER IS OPEN TO ASSURE THAT CONTACT IS NOT MADE WITH ANY OF THESE COMPONENTS.

WARNING: POTENTIAL MAGNET QUENCH

DO NOT attempt to measure resistance by connecting Ohmmeter across magnet heater connections.

DO NOT press the Rundown/Quench button throughout this process.

Do not reconnect MRU cable if it is found disconnected. GEHC strongly recommends that you stop using the system, and immediately call your qualified service representative.

7.1. Quarterly, General Electric Field Service

7.1.1. Examine the exterior of the MRU for the GE "Maintenance or Inspection" label.

7.1.2. Replace the batteries if the date is equal to or greater than 3 years.

7.1.3. Order the batteries for installation during the next PM if the date will be equal to or greater than 3 years at the time of the next scheduled PM event.

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7.2. Yearly, General Electric Field Service

7.2.1. Perform tests in Section 3.

7.2.2. Remove heater connector P2 from J2.

WARNING: ENSURE THE HEATERS ARE NOT CONNECTED TO J2 TO PREVENT INADVERTENT MAGNET QUENCH.

7.2.3. Open the front cover and visually inspect the entire unit for any evidence of overheating, discoloration, If new batteries or lugs are needed, apply a thin layer of silicone grease or petroleum jelly to terminals after assembly.

7.2.4. Reinstall battery hold down plate and fuse F2, 2.5A. Locate 20 ohm 5% 20 watt ceramic resistor in plastic bag (part number 5196919-7), remove and connect to lugs 1 and 2 of TB2. Connect the DVM across this resistor and press the RUN DOWN push-button. The DVM should read approximately 20 volts for approximately 50 seconds. The test resistor must remain energized for a minimum of 30 seconds. Allow a minimum of 3 minutes for switch S4 to cool before subsequent test resistor timing cycles.

7.2.5. Move the test resistor from lugs 1 and 2 of TB2 to lugs 3 and 4 of TB2. Connect the DVM across this resistor and press the RUN DOWN push-button. The DVM should read approximately 20 volts for approximately 50 seconds. The test resistor must remain on for a minimum of 30 seconds. Remove the test resistor from TB2.

7.2.6. Connect the DVM to TP +20.7V and TP ground. The meter should read 20.7V +/- 0.1V. If not, remove fuse F2 and adjust R3 for 20.7V and then reinstall F2.

WARNING: BEFORE CONNECTING P2 TO J2, ASSURE THAT AC POWER HAS BEEN APPLIED TO THE MRU FOR AT LEAST 3 MINUTES AND THAT THE RUN DOWN BUTTON HAS NOT BEEN ACTUATED FOR AT LEAST 3 MINUTES. FAILURE TO HEED THIS WARNING MAY CAUSE A MAGNET QUENCH.

7.2.7. Disconnect the DVM, plug P2 into J2, close the cover, and perform tests 3.1.1 through 3.1.4.

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7.3. Battery Replacement — Applies if battery failure occurs, qualified service representative

7.3.1. Replace all three 6V sealed lead-acid batteries (see replacement parts list). OBSERVE BATTERY POLARITY. Batteries contain lead, and must be disposed of properly.

7.3.2. If the batteries have been stored for one year or more, it may be necessary to wait for as long as 8 hours for battery voltage to reach 20.0 V minimum. Initial charge current is approximately 800 mA which will decrease to approximately 3 mA when fully charged.

7.3.3. If new batteries or lugs are needed, apply a thin layer of silicone grease or petroleum jelly to terminals after assembly.

7.3.4. After battery replacement, obtain a GE "Maintenance or Inspection" label (GE Part number 5661793).

7.3.5. Complete all label entries including the date on which the battery was replaced. See example and use ink pen or permanent marker.

7.3.6. Place label on the exterior, outside bottom of the MRU or in a location that can be accessed easily during a future PM. Can be installed over old label.

7.3.7. Perform the Yearly tests and check battery charger voltage as described in sections of 7.2.

GE MAINTENANCE or INSPECTION

DATE (mm-dd-yy) 06-16-2015

COMPONENT MRU Battery

and COMMENTS

EMPLOYEE ID and INITIALS Name with SSO number

Made by Clarion for GE HealthCare GEHC Part Number 5661793

Example only

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